Pandas I: The Headless Excel

What is Pandas?

- Great addition to the Python stack
- Allows manipulation of structured data with many variables
- Handles time series, missing values, and multiple datasets
- Provides rich data types and operations on DataFrames

Play with Example Notebook

Pandas Data Structures

- 1D array: **Series**
- 2D collection of Series: **DataFrame**

Series Data Structures

Series

```python
# Creating a Pandas Series
s = pd.Series([1, 3, 5, None, 6, 8])
s
```

	0
0	1.0
1	3.0
2	5.0
3	NaN
4	6.0
5	8.0

dtype: float64

Series

```
[50] # next example series with text data

     series_text = pd.Series(['apple', 'banana', 'cherry'])
     series_text
```

	0
0	apple
1	banana
2	cherry

dtype: object

```
[21] df_episode = pd.read_csv(csv_file)
     df_episode
```

	No.	Title	Directed by	Written by	Original release date	Description
0	1	"Gram Panchayat Phulera"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020 (2020-04-03)	Abhishek Tripathi is a fresh engineering gradu...
1	2	"Bhootha Ped"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020 (2020-04-03)	Abhishek, working as panchayat secretary, is a...
2	3	"Chakke Wali Kursi"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020 (2020-04-03)	Abhishek buys a comfy revolving chair since wo...
3	4	"Hamara Neta Kaisa Ho?"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020 (2020-04-03)	Abhishek is instructed by the BDO to put up fa...

Figure: simple Dataframe

Creating a DataFrame

- Use `pd.DataFrame(...)` to create a DataFrame
- Each column can have a different data type

```
[72] # Creating a DataFrame from a dictionary
     data = {'col1': [1, 2, 3], 'col2': ['A', 'B', 'C']}
     df = pd.DataFrame(data)
     df
```

	col1	col2
0	1	A
1	2	B
2	3	C

`df.loc[]` vs. `df.iloc[]`

Left-most column is used as index by default

```
[74] df.loc[0] # label-based (uses row/column names)
```

	0
col1	1
col2	A

dtype: object

```
[75] df.iloc[0] #position-based (uses row/column index)
```

	0
col1	1
col2	A

dtype: object

`df.loc[]` vs. `df.iloc[]`

- `df.loc[]` is **label-based** (uses index names or column names)
- `df.iloc[]` is **position-based** (uses integer positions)

Examples:

- `df.loc[2]` → row with index label 2
- `df.iloc[2]` → 3rd row (0-based)
- `df.loc[2, 'fruit']` → value in row with index 2, column 'fruit'
- `df.iloc[2, 1]` → value in 3rd row, 2nd column

Descriptive Statistics

- Use `df.describe()` for numeric columns
- Summarize categorical values as well

```
[76] df.describe()
```

	col1
count	3.0
mean	2.0
std	1.0
min	1.0
25%	1.5
50%	2.0
75%	2.5
max	3.0

Loading Data from CSV

- `df = pd.read_csv('file.csv')`
- Alternatives:
  - Upload to Github
  - Mount Google Drive
  - Use wget + direct link

```
[21] df_episode = pd.read_csv(csv_file)
     df_episode
```

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Viewing Rows

- `df.head()` shows first 5 rows
- `df.tail()` shows last 5 rows

```
[81] display(df.head())
     display(df.tail())
```

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3	4	"Hamara Neta Kaisa Ho?"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020 (2020-04-03)	Abhishek is instructed by the BDO to put up fa...
4	5	"Computer Nahi Monitor"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020 (2020-04-03)	Frustrated by the lack of nightlife in the vil...

	No.	Title	Directed by	Written by	Original release date	Description
19	4	"Atma Manthan"	Deepak Kumar Mishra	Chandan Kumar	28 May 2024 (2024-05-28)	The final blow to Pradhan's reputation comes a...
20	5	"Shanti Samjhauta"	Deepak Kumar Mishra	Chandan Kumar	28 May 2024 (2024-05-28)	Manju Devi reluctantly agrees to Bhushan's pro...
21	6	"Chingaari"	Deepak Kumar Mishra	Chandan Kumar	28 May 2024 (2024-05-28)	To fix their dented image, the Pradhan gang se...
22	7	"Shola"	Deepak Kumar Mishra	Chandan Kumar	28 May 2024 (2024-05-28)	The MLA puts out a hit on one of the village r...
23	8	"Hamla"	Deepak Kumar Mishra	Chandan Kumar	28 May 2024 (2024-05-28)	Finally, Pradhan rallies the whole Phulera beh...

Slicing a DataFrame

- Use `df[rows:columns]` syntax
- Combine loc and iloc for powerful access

```
[84] # SLICING
     df.loc[2:5,'Description']
```

	Description
2	Abhishek buys a comfy revolving chair since wo...
3	Abhishek is instructed by the BDO to put up fa...
4	Frustrated by the lack of nightlife in the vil...
5	Abhishek is having his photograph taken for th...

dtype: object

```
[86] # SLICING
     df.iloc[0:2,1]
```

	Directed by
0	Deepak Kumar Mishra
1	Deepak Kumar Mishra

dtype: object

Conditional Filtering

- Filter rows using conditions: `df[df["col"] > 20]`
- Chain conditions: `df[(cond1)`
  `(cond2)]`

```
[44] df[(df['sales_kg'] > 20) & (df['fruit'] == 'apple')]
```

	date	fruit	sales_kg	price_per_kg	shop
0	2025-01-01	apple	23	2.5	Bagar
3	2025-01-04	apple	34	2.5	Bagar
6	2025-01-07	apple	23	2.5	Bagar
9	2025-01-10	apple	34	2.5	Bagar
10	2025-01-11	apple	34	2.5	Bagar
11	2025-01-11	apple	34	2.5	Bagar

Sorting and Indexing

- `df.sort_index()` sorts by index
- DataFrame = 2D matrix with axis 0 (rows), axis 1 (columns)

```
[92] display(df2)
     df2.sort_index()
```

	animal	fruit
food		
dal	ant	grapes
bhat	cow	apple
tarkari	dog	banana

	animal	fruit
food		
bhat	cow	apple
dal	ant	grapes
tarkari	dog	banana

Adding and Dropping Rows/Columns

- Drop rows: `df.drop(['row1', 'row2'])`
- Drop columns: `df.drop('col1', axis=1)`

```
[60] df.drop([0, 2]) # Drops rows with index 0 and 2
     #  permanently drop with (inplace=True)
```

```
[61] df.drop('price_per_kg', axis=1)#drop the 'price_per_kg' column

     #  permanently drop with (inplace=True)
```

Removing Duplicates

- Use `df.drop_duplicates(subset='col')`
- Use `df['col'].value_counts()` for frequency

```
[47] df_no_dups = df.drop_duplicates() # see exact matches only
     df_no_dups
```

```
[57] df['fruit'].value_counts() #  how many times each unique fruit appears in the 'fruit' column?
```

Groupby and Aggregation

- Use `df.groupby("col").agg(["mean", "sum"])`
- Apply custom aggregation functions too

```python
# summarizes each fruit's total and average sales, average price, and total revenue
df.groupby('fruit').agg({
    'sales_kg': ['sum', 'mean'],
    'price_per_kg': 'mean',
    'total': 'sum'
})
```

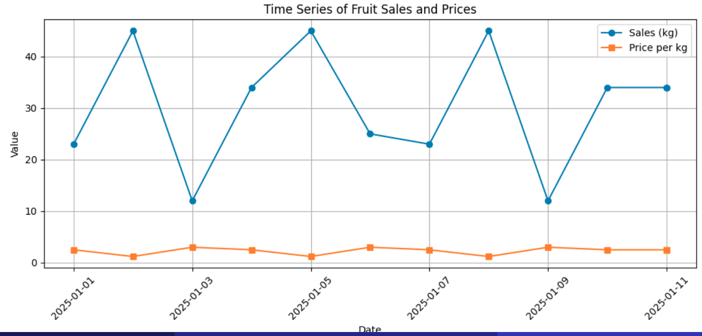
	sales_kg		price_per_kg	total
	sum	mean	mean	sum
fruit				
apple	182	30.333333	2.5	455.0
banana	135	45.000000	1.2	162.0
orange	49	16.333333	3.0	147.0

Plotting with Pandas

- Simple visualization: `df.plot()`
- Use built-in plotting methods for quick insights

```
[49] plt.figure(figsize=(10, 5))
     plt.plot(df_time.index, df_time['sales_kg'], marker='o', label='Sales (kg)')
     plt.plot(df_time.index, df_time['price_per_kg'], marker='s', label='Price per kg')

     plt.title('Time Series of Fruit Sales and Prices')
     plt.xlabel('Date')
     plt.ylabel('Value')
     plt.legend()
     plt.grid(True)
     plt.xticks(rotation=45)
     plt.tight_layout()
     plt.show()
```



References

- McKinney, Wes. *Python for Data Analysis*. O'Reilly Media, 2nd Edition, 2018.
- CS540: Database & Information Retrieval Lecture Notes by Dr. Yongxin (Jack) Liu, ERAU Daytona Beach
- Official Pandas Documentation: https://pandas.pydata.org/docs/

Questions?

Thank You!